Practical No. 6

Problem Statement : Write and execute subqueries to retrieve data from one table based on results from another.

StudentDb

```
CREATE DATABASE StudentDb;
USE StudentDb;
CREATE TABLE Student_Info (
NAME VARCHAR(50),
ROLL_NO INT PRIMARY KEY,
 LOCATION VARCHAR(50),
 PHONE_NUMBER VARCHAR(15)
);

CREATE TABLE Student_Section (
NAME VARCHAR(50),
 ROLL_NO INT,
 SECTION CHAR(1)
);

INSERT INTO Student_Info VALUES
(‘Hetal’, 101, 'Pune”, '1111111111'),
(‘Jiya’, 102, 'Mumbai', '2222222222'),
(‘Neha’, 103, 'London', '3333333333'),
(‘Sita’, 104, 'Berlin', '4444444444'),
(‘John’, 105, 'Tokyo', '5555555555');

INSERT INTO Student_Section VALUES
(‘Sara’, 104, 'A'),
```

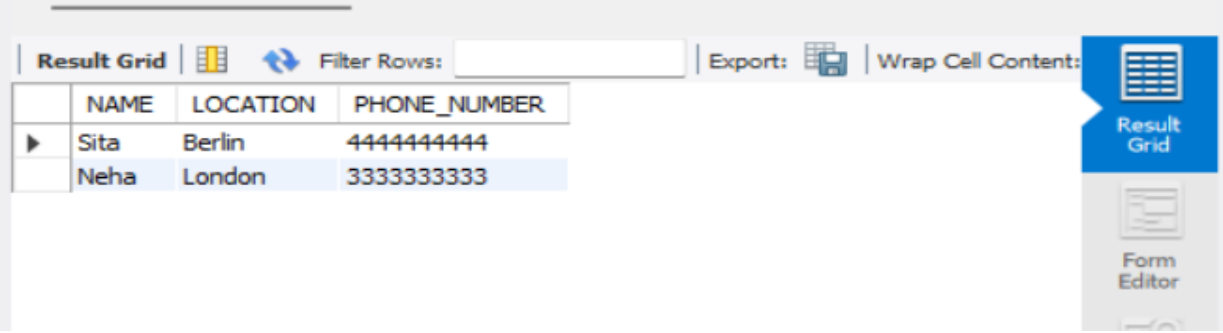
(‘Ram’,102, 'B'),

(‘Vinay’,103, 'A');

Example 1: Subquery in WHERE Clause

```
SELECT NAME, LOCATION, PHONE_NUMBER
FROM Student_Info
WHERE ROLL_NO IN (
SELECT ROLL_NO
FROM Student_Section
WHERE SECTION = 'A'
);
```

Output :-



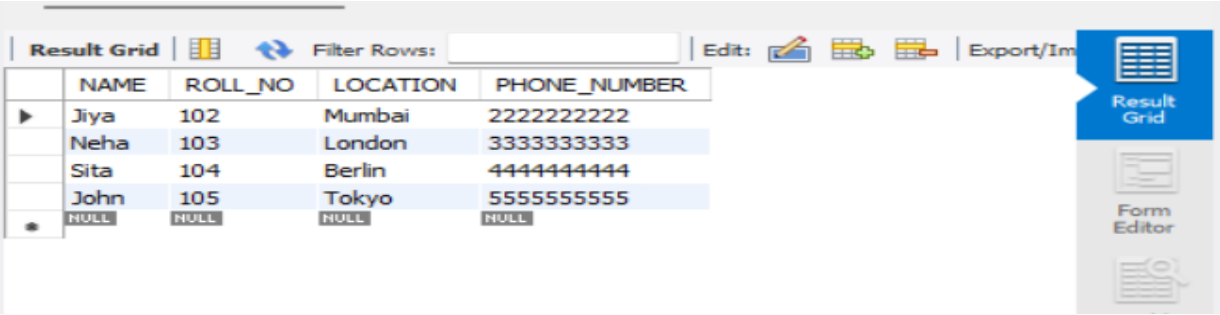
NAME	LOCATION	PHONE_NUMBER
Sita	Berlin	4444444444
Neha	London	3333333333

Example 2: Subquery with DELETE

```
SET SQL_SAFE_UPDATES=0;
DELETE FROM Student_Info
WHERE ROLL_NO <= 101 OR ROLL_NO = 201;

SELECT * FROM Student_Info;
```

Output:



NAME	ROLL_NO	LOCATION	PHONE_NUMBER
Jiya	102	Mumbai	2222222222
Neha	103	London	3333333333
Sita	104	Berlin	4444444444
John	105	Tokyo	5555555555
NULL	NULL	NULL	NULL

Example 3: Using Subquery with UPDATE

Query:

```
UPDATE Student_Info
SET NAME = 'amit'
WHERE LOCATION IN ('London', 'Berlin');

SELECT * FROM Student_Info;
```

Output :-

Result Grid | Filter Rows: | Edit: | Export/Im

NAME	ROLL_NO	LOCATION	PHONE_NUMBER
Jiya	102	Mumbai	2222222222
amit	103	London	3333333333
amit	104	Berlin	4444444444
John	105	Tokyo	5555555555
NULL	NULL	NULL	NULL

Result Grid | Form Editor | Field Types

Example 4: Simple Subquery in the FROM Clause

Query:

```
SELECT NAME, PHONE_NUMBER
FROM (
   SELECT NAME, PHONE_NUMBER, LOCATION
   FROM Student_Info
   WHERE LOCATION LIKE 'T%'
) AS subquery_table;
```

Output :-

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

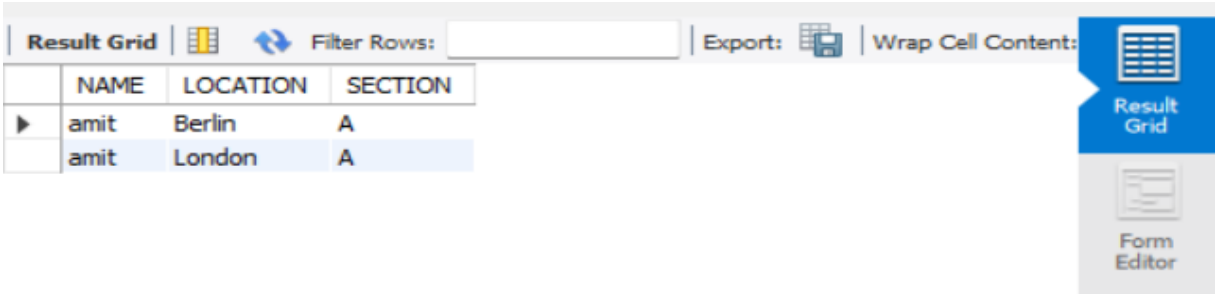
NAME	PHONE_NUMBER
John	5555555555

Result Grid | Form Editor

Example 5: Subquery with JOIN
Query :-

```
SELECT s.NAME, s.LOCATION, ns.SECTION
FROM Student_Info s
INNER JOIN (
   SELECT ROLL_NO, SECTION
   FROM Student_Section
   WHERE SECTION = 'A'
) ns
ON s.ROLL_NO = ns.ROLL_NO;
```

Output:-



NAME	LOCATION	SECTION
amit	Berlin	A
amit	London	A